

Lab: Research Frontiers and Course Capstone

Learning Goal: Synthesize all 401 course content into a research proposal and demonstrate mastery of Active Inference research methods.

Part 1: Frontier Analysis

Exercise: For each frontier, assess the current state and identify the most impactful next experiment:

Frontier	Maturity (1-10)	Strongest Evidence	Biggest Gap	Next Key Experiment
Neuroscience (brain implementation)	7	DCM connectivity studies, MMN precision effects	Laminar-level validation of canonical microcircuit	Multi-electrode laminar recordings during prediction paradigm
Computational psychiatry	5	Schizophrenia precision models, depression computational phenotyping	Lack of treatment trials using computational enrichment	RCT with computational phenotype-based patient selection
AI and robotics	4	pymdp/RxInfer implementations, simple robot demos	Scalability to real-world complexity	Active Inference agent competing on standard RL benchmarks
Consciousness	3	Self-modeling theory, depth predictions	No operational test distinguishing AI predictions	Paradigm generating consciousness- specific Active Inference prediction
Biology/ morphogenesis	2	Theoretical papers on FEP and development	Almost no empirical biological validation	In vitro cell system testing FEP morphogenesis predictions

Choose one frontier and write a 200-word research vision.

Write your response here

Part 2: Research Proposal

Learning Goal: Write the Specific Aims page for a research proposal.

Exercise: Write a 1-page Specific Aims for an Active Inference research proposal:

Template:

Title: [Your title]

Background: [2-3 sentences on the problem and significance]

Overall objective: [1 sentence stating the overarching goal]

Central hypothesis: [1 sentence stating the key hypothesis, grounded in Active Inference]

Rationale: [2-3 sentences explaining why Active Inference is the right framework and what evidence supports feasibility]

Specific Aim 1: [Title] — [1-2 sentences describing the aim, approach, and expected outcome]

Specific Aim 2: [Title] — [1-2 sentences]

Specific Aim 3: [Title] — [1-2 sentences]

Expected outcomes and impact: [2-3 sentences on what success looks like and why it matters]

Write your response here

Part 3: Methods Integration

Learning Goal: Demonstrate ability to combine methods from multiple course modules.

Exercise: Design a study that uses methods from at least 4 of the 8 Research Methods modules:

Module Used	Method	Its Role in Your Study
01 (DCM)	Dynamic Causal Modelling	Estimate effective connectivity changes
02 (Phenotyping)	Computational phenotyping	Characterize individual computational profiles
03 (Electrophysiology)	EEG/MMN	Measure precision-weighted prediction errors
04 (Behavioral)	Reversal learning task	Dissociate learning rate and volatility estimates
05 (Robotics)	pymdp simulation	Generate model predictions for comparison
06 (Clinical)	Clinical translation	Apply to patient population
07 (Interdisciplinary)	Open science practices	Pre-register, share data, open-source code
08 (Frontiers)	Novel frontier	Apply to emerging domain

Describe how these methods integrate into a coherent study design.

Write your response here

Part 4: Critical Evaluation

Learning Goal: Demonstrate ability to critically evaluate Active Inference research.

Exercise: Read the following claim and write a 300-word critical evaluation:

Claim: "Active Inference will replace reinforcement learning as the dominant framework for artificial intelligence because it provides a mathematically principled account of exploration, model-based reasoning, and multi-agent coordination that RL lacks."

Evaluate by addressing: (a) In what respects is this claim supported? (b) In what respects is it overconfident? (c) What would need to be demonstrated empirically for this claim to be credible? (d) What are RL's genuine advantages that Active Inference currently lacks?

Write your response here

Part 5: Course Capstone Reflection

Exercise: Write a 500-word essay reflecting on the entire 401 course. Address:

1. **What is Active Inference?** After all four tracks, how do you define it — a theory, framework, principle, mathematical language, research program, or something else?
2. **What have you learned?** What is the single most important insight from each track?

3. Philosophical Foundations: _____

4. Neuroscientific Frontiers: _____

5. Advanced Theory: _____

6. Research Methods: _____

7. **What are the limits?** What are Active Inference's three biggest weaknesses or unsolved problems?

8. **What will you do next?** How will you apply what you've learned? What research question will you pursue?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Frontier analysis	Research landscape assessment
2	Proposal writing	Specific Aims development
3	Methods integration	Multi-modal study design
4	Critical evaluation	Balanced assessment of claims
5	Course synthesis	401 capstone reflection